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VEHICLE SEAT WITH A LONGITUDINAL ADJUSTMENT LOCK

Abstract

A vehicle seat comprises a longitudinal adjustment device with a first adjustment part which can be adjusted relative to a second adjustment part along a longitudinal direction, a locking device by means of which the first adjustment part can be arrested relative to the second adjustment part, and an operating part movably mounted on the first adjustment part, by actuation of which the locking device can be unlocked, a seat part which is supported on the second adjustment part via the first adjustment part, and an adjustment lock with a locking element mounted on the operating part and a mating locking element fixed to the second adjustment part, wherein the locking element can be locked to the mating locking element in at least one position of the first adjustment part relative to the second adjustment part.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a U.S. National Phase of International Application No. PCT/EP2023/060563 entitled "VEHICLE SEAT WITH A LONGITUDINAL ADJUSTMENT LOCK," and filed on Apr. 24, 2023. International Application No. PCT/EP2023/060563 claims priority to German Patent Application No. 10 2022 204 027.9 filed on Apr. 26, 2022. The entire contents of each of the above-listed applications are hereby incorporated by reference for all purposes.

BACKGROUND
[0002] This solution relates to a vehicle seat.
[0003] In dependence on the respective site of use, i.e. in particular the shape and size of a door opening of a vehicle with the vehicle seat, it may be desirable to clear an entry region as large as possible. What is conceivable therefor are vehicle seats which combine e.g. folding forward of a backrest with unlocking of a longitudinal adjustment device in order to bring the vehicle seat into an easy-entry position. However, as a result of its weight force, the vehicle seat in such cases often is automatically moved back from the easy-entry position in the direction of a position of use, e.g. when the vehicle is standing uphill. This can make entry more difficult and can also involve a risk of injury.
[0004] DE 10 2020 205 108 A1 describes a vehicle seat with a locking portion integrated on a kinematic lever for comfortably locking and unlocking the vehicle seat in the easy-entry position.
[0005] A vehicle seat which is longitudinally shiftable by tilting its backrest is described in DE 36 08 827 A1. The function of this vehicle seat, however, is based on a relatively complex lever construction.
[0006] It is and object of the underlying the proposed solution to indicate a vehicle seat which has as simple a construction as possible.
[0007] This object is achieved by a vehicle seat having features as described herein.
[0008] Accordingly, a vehicle seat comprises a longitudinal adjustment device with a first adjustment part, which is adjustable relative to a second adjustment part along a longitudinal direction, a locking device by means of which the first adjustment part can be arrested relative to the second adjustment part, and an operating part movably mounted on the first adjustment part, by actuation of which the locking device can be unlocked, and a seat part which is supported on the second adjustment part via the first adjustment part. Furthermore, the vehicle seat comprises an adjustment lock with a locking element mounted on the operating part and a mating locking element fixed to the second adjustment part. In at least one position of the first adjustment part relative to the second adjustment part, the locking element can be locked to the mating locking element.
[0009] This construction allows locking e.g. in a forward position of the vehicle seat with particularly few additional parts, which due to the simple construction can also be of the easily manufacturable type, so that a vehicle seat of particularly simple construction becomes possible.
[0010] For example, locking of the locking element to the mating locking element can be released by actuating the operating part. This allows a particularly intuitive operation. The operating part also obtains a dual function which contributes to the simple construction.
[0011] The vehicle seat can comprise a kinematic device by means of which the seat part is movable between at least one position of use to be occupied and a position moved forwards along

the longitudinal direction. The kinematic device optionally connects the seat part to the longitudinal adjustment device. The kinematic device can comprise one or more pivot levers. Such a kinematic device provides for an easy-entry position moved forwards particularly far for a particularly comfortable entry.
[0012] The kinematic device can be locked e.g. in the position of use. An inadvertent adjustment of the seat part can thereby be prevented.
[0013] Optionally, the vehicle seat comprises an unlocking mechanism, by actuation of which the kinematic device and/or the longitudinal adjustment device, in particular the kinematic device and the longitudinal adjustment device jointly, can be unlocked. This provides for a particularly intuitive and easy operation.
[0014] The kinematic device can include a multiple joint. For example, the kinematic device forms a four-bar linkage or a seven-bar linkage. The kinematic device for example can comprise at least two pivot levers, which each are formed to be pivotable with the seat part and the first adjustment part.
[0015] Optionally, the locking element is formed spring-elastic. For locking to the mating locking element, the locking element can snap into place thereon in a simple way. This provides for a simple and at the same time particularly robust configuration.
[0016] For example, the locking element is configured in the form of a leaf spring. Such a locking element can be manufactured particularly easily.
[0017] Furthermore, the mating locking element can include an insertion bevel for the locking element. The e.g. spring-elastic locking element can slide along the same in a particularly simple way for locking. Alternatively, it can also be provided that the locking element is rigid and/or includes an insertion bevel and that the mating locking element is of spring-elastic design.
[0018] Optionally, the locking element includes an eyelet. It can be provided that the mating locking element can lockingly engage into the eyelet. This provides for a particularly robust and safe engagement of the mating locking element.
[0019] The locking element can be attached to an arm of the operating part, in particular to an arm by means of which the locking device of the longitudinal adjustment device can be unlocked. This provides for a further simplification of the construction.
[0020] The vehicle seat furthermore can include an unlocking element pivotally mounted on the first adjustment part, via which the arm is operatively connected to the locking device of the longitudinal adjustment device. This provides for an effective power transmission.
[0021] The operating part in particular can include a manually actuatable handle portion. The handle portion can be of rod-shaped design. This allows a particularly easy manufacture and intuitive operation.
[0022] The operating part is e.g. pivotally mounted on the first adjustment part. This provides for a particularly simple mechanism.
[0023] Optionally, the operating part is pretensioned by means of a spring into a position in which the locking element can be locked to the mating locking element. It can thus be ensured that locking is effected in the easy-entry position.
[0024] The adjustment parts can be movable relative to each other along an adjustment path, wherein the locking element and the mating locking element are configured to lock the adjustment parts to each other at an end of the adjustment path. The vehicle seat thereby can be comfortably held in the end position. The adjustment parts in particular can be configured in the form of an upper rail and a lower rail. This provides for an easy-entry position moved forwards particularly far, which nevertheless snaps into place comfortably and by using few parts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The idea underlying the solution will be explained in detail below with reference to the exemplary embodiment illustrated in the Figures. In schematic representations:

[0026] FIG. **1** shows a view of a vehicle seat with a seat part and a backrest.

[0027] FIG. **2** shows a view of a longitudinal adjustment device of the vehicle seat with an adjustment lock.

[0028] FIGS. **3-7** show views of a locking element and a mating locking element of the adjustment lock in various positions of the vehicle seat.

DETAILED DESCRIPTION

[0029] A vehicle seat **1** schematically shown in FIG. **1** includes a seat part **10** and a backrest part **15** with a head rest **16** pivotally arranged with respect to the seat part **10** via an arrangement of rotary fittings **14**. A seat user can sit down on the seat part **10**. Via the arrangement of rotary fittings **14** the backrest part **15** can be pivoted relative to the seat part **10** in order to adapt the tilt position of the backrest part **15** with respect to the seat part **10** and/or bring the backrest part **15** into a position pivoted forwards.

[0030] Such a vehicle seat **1** can be configured as a front seat in a vehicle. Such a vehicle seat **1** can, however, also be used e.g. as a rear seat, for example in a second seat row, in a vehicle.

[0031] In the illustrated example, the seat part **10** of the vehicle seat **1** is coupled to a longitudinal adjustment device **13** via a kinematic device **12**. The longitudinal adjustment device **13** carries the seat part **10**. As in the illustrated example the backrest part **15** is mounted on the seat part **10**, the longitudinal adjustment device **13** thus also carries the backrest part **15**. A weight force of the seat part **10** (and of the backrest part **15**) and, if a seat user sits down on the vehicle seat **1**, also a weight force of the seat user, hence is introduced into the longitudinal adjustment device **13** via the kinematic device **12**.

[0032] Via the longitudinal adjustment device **13**, the seat part **10** (jointly with the kinematic device **12** and the backrest part **15**) can be connected to a vehicle floor FB of a vehicle and is connected to the same according to FIG. **1** in a longitudinally adjustable way. The longitudinal adjustment device **13** comprises a first adjustment part, here in the form of an upper rail **130** which is shiftably mounted on a second adjustment part, here in the form of a lower rail **131**, along a longitudinal direction X. The lower rail **131** can be firmly connected to the vehicle floor FB and is firmly connected to the same according to FIG. **1**. In the present case, the vehicle seat **1** comprises two of such rail pairs, one each on one side of the vehicle seat **1**. By means of the longitudinal adjustment device **13** the vehicle seat is adjustable in the longitudinal direction X.

[0033] The longitudinal adjustment device **13** comprises a locking device **133** by means of which the upper rail **130** can be arrested relative to the lower rail **131**, namely can be locked to the lower rail **131**.

[0034] In the present case, an adjustment part, namely the upper rail **130**, of the longitudinal adjustment device **13** jointly serves as a base for the kinematic device **12**.

[0035] A locking device **122** of the kinematic device **12** secures the vehicle seat **1** in a position of use in which the seat user can sit down on the seat part **10**. When this locking device **122** is released, kinematic levers **120**, **121** of the kinematic device **12** can be pivoted relative to the upper rail **130** so that the vehicle seat **1** can be transferred from the position of use into a position moved forwards with respect thereto in order to provide an entry position. Via the kinematic levers **120**, **121** the seat part **10** is connected to the upper rail **130** so as to be movable relative to the upper rail **130**. Optionally, the kinematic device **12** also serves for adjusting the seat height and allows the setting of various positions of use.

[0036] As will yet be explained in detail below, the vehicle seat **1** can be transferred into an entry position by a pivotal movement by means of the kinematic device **12** and by a shifting movement by means of the longitudinal adjustment device **13**. The entry position can also be referred to as an easy-entry position.

[0037] The vehicle seat **1** comprises an unlocking mechanism **17**, which here is illustrated by way of example as an unlocking lever. An actuation of the unlocking mechanism **17** effects unlocking of the locking device **122** of the kinematic device **12** and of the locking device **133** of the longitudinal adjustment device **13**. In the illustrated example a Bowden cable is provided, which can be pulled through the unlocking mechanism **17** in order to unlock both locking devices **122**, **133**. Then, the vehicle seat **1** can be pivoted and shifted forwards.

[0038] FIG. **2** shows two rails **130**, **131** of the longitudinal adjustment device **13**. At this point, it should be noted that FIG. **2** shows one of two sides of the vehicle seat **1**, wherein the vehicle seat **1** comprises a second side, in particular a mirror-symmetrically constructed second side. The description of the side shown in FIG. **2** analogously applies for the second side of the vehicle seat **1** shown in FIG. **1**.

[0039] The first kinematic lever **120** shown in FIG. **1**, i.e. the rear kinematic lever **120** with respect to the longitudinal direction X, is pivotally mounted on the upper rail **130** on a rear swivel bearing S**1**. The second kinematic lever **121**, i.e. the front kinematic lever **120** with respect to the longitudinal direction X, is mounted on the upper rail **130** so as to be pivotable about a front swivel bearing S**2**. Furthermore, the kinematic levers **120**, **121** each are pivotally mounted on the seat part **10**. The upper rail, the two kinematic levers **120**, **121** and the seat part **10** jointly form a four-bar linkage.

[0040] The four-bar linkage of the kinematic device **12** is configured such that during the forward and backward displacement between the position of use and the easy-entry position the seat part **10** is first lifted and then lowered again. Therefore, the kinematic device **12** can also be referred to as “bunny hop kinematics”. This provides for a far displacement forwards into the easy-entry position. In the present case, this movement is supported by a pretensioned spring **123** which pretensions the seat part **10** in the direction of the easy-entry position. The spring **123** is configured in the form of a coil spring extended about a pivot axis defined by the front swivel bearing S**2**.

[0041] In the illustrated example, the locking device **133** of the longitudinal adjustment device **13** includes a comb which is movably, e.g. pivotally mounted on the upper rail **130** and can lockingly engage into holes at the lower rail **131** (see e.g. FIG. **1**). The locking device **133** can be unlocked by means of a bolt **137**. In the present case, the bolt **137** therefor can be displaced in a slot in the upper rail **130** so that the comb is lifted out of the holes and enables a relative movement of the rails **130**, **131**.

[0042] For unlocking the locking device **133** an unlocking element **139** is provided. The unlocking element **139** is pivotally mounted on the upper rail **130** at a pivot axis. A lever arm of the unlocking element **139** is operatively connected to the bolt **137**. A pivotal movement of the unlocking element **139** lifts the bolt **137** so that the locking of the rails **130**, **131** is released. The unlocking element **139** furthermore has a further lever arm.

[0043] To effect a pivotal movement of the unlocking element **139**, the vehicle seat **1** comprises an operating part **132**. The operating part **132** is pivotally mounted on the upper rail **130**. It includes a handle portion **135**. The handle portion **135** is formed by a bent rod and can also be referred to as a “towel bar”. A seat user can actuate the handle portion **135**, more exactly: pull the same upwards. The operating part **132** furthermore comprises an arm **134**. In the present case, the operating part **132** furthermore comprises a rod **138** to which the handle portion **125** and the arm **134** are attached, wherein other configurations also are conceivable. The rod **138** is straight. Via the rod **138**, the handle portion **135** and the arm **134** are firmly connected to each other and pivotally mounted on the upper rail **130**. A spring **136** is spring-elastically pretensioned and urges the operating part **132** against the unlocking position.

[0044] The arm **134** of the operating part **132** cooperates with the unlocking element **139**, namely in the present case with the further lever arm of the unlocking element **139**. When the arm **134** of the operating part **132** pivots relative to the upper rail **130**, the arm **134** presses against the further lever arm of the unlocking element **139** so that the same is pivoted.

[0045] The vehicle seat **1** furthermore comprises an adjustment lock **11** with a locking element **110** mounted on the operating part **132** and a mating locking element **111** attached to the second adjustment part, here to the lower rail **131**, wherein the locking element **110** can be locked to the mating locking element **111** in at least one position of the first adjustment part, i.e. here of the upper rail **130**, relative to the second adjustment part, i.e. to the lower rail **131**.

[0046] The locking element **110** in the present case is mounted on the arm **134** of the operating part **132**. The locking element **110** is configured in the form of a spring, namely in the illustrated example in the form of a leaf spring, wherein a coil spring or another type of spring would also be conceivable. The locking element **110** flatly rests on the arm **134**. It is attached to the arm **134** in the region of an end of the arm **134**. At the opposite end, the locking element **110** includes an eyelet **113**. In the region of the eyelet **113** the locking element **110** is bent, namely in the present case is U-shaped.

[0047] The mating locking element **111** is attached to the lower rail **131**. In the present case, it is bent in an S-shaped manner. The mating locking element **111** has an open end. This open end can be locked to the locking element **110**. The mating locking element **111** therefor includes an insertion bevel **112** along which the locking element **110** can slide during a movement of the upper rail **130** relative to the lower rail **131**. The locking element is bent spring-elastically. During a movement into the easy-entry position the eyelet **113** is lifted when sliding along the insertion bevel **112** and snaps back as soon as it is pushed over the open end of the mating locking element **111**. The eyelet **113** thereby comes into locking engagement with the mating locking element **111**. The locking element **110** and the mating locking element **111** are configured to lock the rails **130**, **131** to each other at the end of the adjustment path.

[0048] This position is shown in FIGS. **3** and **4**. The eyelet **113** is in engagement with the mating locking element **111**, so that a displacement of the upper rail **130** relative to the lower rail **131** back towards the position of use is blocked. The eyelet **113** abuts against the open end of the mating locking element **111** and thus prevents a further movement.

[0049] When the vehicle seat **1** now is to be displaced again from the forward position, the easy-entry position, back into the position of use, it is sufficient to actuate the operating part **132**. This can be effected manually at the handle portion **135**. Optionally, the unlocking mechanism **17** is configured to pivot the operating part **132**.

[0050] FIG. **5** shows the operating part **132** in a raised position as compared to FIGS. **3** and **4**. Furthermore, the position of the locking element **110** is shown with this position of the operating part **132**. The locking element **110**, concretely the eyelet **113** of the locking element **110**, is lifted from the mating locking element **111** and out of engagement with the same.

[0051] Due to this unlocking, the upper rail **130** with the part of the vehicle seat **1** mounted thereon can be shifted relative to the lower rail **131** from the forward position back to the position of use.

[0052] FIG. **6** shows a position in which the upper rail **130** with the part of the vehicle seat **1** mounted thereon already has been shifted relative to the lower rail **131** to a certain extent from the forward position back towards the position of use.

[0053] Thus, a locking of the locking element **110** to the mating locking element **111** can be released in a simple way by actuating the operating part **132**.

[0054] FIG. **7** finally shows the position during a renewed displacement of the vehicle seat **1** forwards into the easy-entry position, in which the locking element **110** impinges on the insertion bevel **112** of the mating locking element **111** when the operating part **132** is not actuated. Proceeding therefrom, a further forward displacement of the upper rail **130** effects that the eyelet **113** slides along the insertion bevel **112** and then snaps into place at the rigid mating locking element **111**.

[0055] The operating part **132** is pretensioned by means of the spring **136** into a position in which the locking element **110** can be locked to the mating locking element **111**.

[0056] Due to the described adjustment lock **11**, the number of components can be kept particularly small. Furthermore, the assembly is particularly easy and no additional bearing points are necessary. Painting of individual parts also can be omitted.
[0057] It should be mentioned that alternatively the locking element **110** conversely might be of rigid design, while the mating locking element **111** would be of spring-elastic design, e.g. also in the form of a leaf spring.
LIST OF REFERENCE NUMERALS
1 vehicle seat
10 seat part
11 adjustment lock
110 locking element
111 mating locking element
112 insertion bevel
113 eyelet
12 kinematic device
120 first kinematic lever
121 second kinematic lever
122 locking device
123 spring
13 longitudinal adjustment device
130 first adjustment part (upper rail)
131 second adjustment part (lower rail)
132 operating part
133 locking device
134 arm
135 handle portion
136 spring
137 bolt
138 rod
139 unlocking element
14 rotary fitting arrangement
15 backrest part
16 headrest
17 unlocking mechanism
FB vehicle floor
S**1**, S**2** swivel bearing
X longitudinal direction

Claims

1. A vehicle seat, comprising: a longitudinal adjustment device with a first adjustment part which can be adjusted relative to a second adjustment part along a longitudinal direction, a locking device by means of which the first adjustment part can be arrested relative to the second adjustment part, and an operating part movably mounted on the first adjustment part, by actuation of which the locking device can be unlocked, a seat part which is supported on the second adjustment part via the first adjustment part, and an adjustment lock with a locking element mounted on the operating part and a mating locking element fixed to the second adjustment part, wherein the locking element can be locked to the mating locking element in at least one position of the first adjustment part relative to the second adjustment part.
2. The vehicle seat according to claim 1, wherein a locking of the locking element to the mating

locking element can be released by actuating the operating part.
3. The vehicle seat according to claim 1, further comprising a kinematic device by means of which the seat part is movable between at least one position of use to be occupied and a position moved forwards along the longitudinal direction.
4. The vehicle seat according to claim 3, wherein the kinematic device can be locked in the position of use.
5. The vehicle seat according to claim 4, further comprising an unlocking mechanism by actuation of which the kinematic device and the longitudinal adjustment device can be jointly unlocked.
6. The vehicle seat according to claim 3, wherein the kinematic device forms a multiple joint.
7. The vehicle seat according to claim 1, claims, wherein the locking element is of spring-elastic design.
8. The vehicle seat according to claim 7, wherein the locking element is configured in the form of a leaf spring.
9. The vehicle seat according to claim 1, wherein the mating locking element includes an insertion bevel for the locking element.
10. The vehicle seat according to claim 1, wherein the locking element includes an eyelet into which the mating locking element can lockingly engage.
11. The vehicle seat according to claim 1, wherein the locking element is attached to an arm of the operating part, by means of which the locking device of the longitudinal adjustment device can be unlocked.
12. The vehicle seat according to claim 11, further comprising an unlocking element pivotally mounted on the first adjustment part, via which the arm is operatively connected to the locking device of the longitudinal adjustment device.
13. The vehicle seat according to claim 1, wherein the operating part includes a manually actuatable handle portion.
14. The vehicle seat according to claim 1, wherein the operating part is pivotally mounted on the first adjustment part.
15. The vehicle seat according to claim 1, wherein the operating part is pretensioned by means of a spring into a position in which the locking element can be locked to the mating locking element.
16. The vehicle seat according to claim 1, wherein the adjustment parts are movable relative to each other along an adjustment path, wherein the locking element and the mating locking element are configured to lock the adjustment parts to each other at an end of the adjustment path.